

Kashmith Samaranayaka

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer and accomplished graduate from the **University of Moratuwa**, specializing in **Machine Learning, Data Science, and Software Engineering**. Proven experience in developing production-ready ML systems, with expertise in MLOps, backend development, and research. Passionate about leveraging cutting-edge AI technologies to solve complex real-world problems.

EXPERIENCE

AI/ML Engineer

April 2024 - Present

Jaseci Lab, 5559 Great Hawk Circle, Ann Arbor, MI 48105, USA

Remote

- Developed and maintained backend infrastructure for **Tobu**, a social media platform with integrated LLMs for enhanced user interactions and automated content generation
- Implemented machine learning features for user image analysis and conversation processing using advanced NLP techniques
- Built scalable data pipelines using PySpark for user data analysis and feature optimization across the platform

Data Scientist

April 2024 - June 2024

Utech Technologies (Pvt) Limited, No 683, Negombo Road, Mabole, Wattala, Sri Lanka

Remote

- Designed and deployed end-to-end energy forecasting pipeline using MLflow for experiment tracking and model versioning
- Implemented automated ML workflows with Apache Airflow for data ingestion, model retraining, and deployment processes
- Built production-ready API services using FastAPI and Docker for real-time energy prediction and dashboard visualization

Application Engineer Intern

January 2023 - July 2023

Synopsys Lanka(Pvt) Ltd, Colombo No. 03, 2/1, Lukshmi Gardens Colombo 08, Sri Lanka

On-site

- Developed NLP-based search engine with LSTM models for automated email classification and content analysis
- Automated large-file compilation processes and implemented notification systems using Python, Bash, and CSH scripting
- Managed 17 processor designs through custom automation scripts for daily performance assessments and resource optimization

EDUCATION

B.Sc. Engineering (Hons.) in Electrical Engineering (Washington Accord accredited) Jan. 2020 - July 2024

University of Moratuwa

Moratuwa, Sri Lanka

- **Overall GPA: 3.31 - Second Upper Class**
- Dean's List – 2nd semester

GCE Advanced Level (A/L) Examination

April 2016 - Dec. 2018

H/Debarawewa National School

Sri Lanka

- Stream: Physical science stream (Combined Mathematics, Physics, Chemistry)
- Results: **3A Passes, Z Score: 2.0495**

PROJECTS

Academic Projects

Self-Adaptive Non-Intrusive Load Monitoring System (NILM) using Deep Learning [↗](#)

- Designed a system using Wavenet and CNN models with custom layers and models for appliance power consumption prediction.
- Integrated ensemble methods for improved prediction accuracy.
- Implemented pseudo-labelling for dynamic adaptation to appliance ageing and wear, reducing the need for retraining.
- Explored transfer learning using pre-trained models with fine-tuning for self-adaptation.
- Published a research paper titled "Self-Adaptive Non-Intrusive Load Monitoring Using Deep Learning" at the 2024 IEEE World AI IoT Congress and won the best-presented paper award.

Industry Projects

Tobu - Social Media Application [↗](#)

- Developed the backend for a social media app integrated with large language models (LLMs) to enhance user interaction and content recommendation.
- Implemented features for user images and conversation analysis using machine learning techniques.
- Worked with LLMs to automate content creation and leveraged PySpark to build data pipelines for user data analysis and feature optimization.

Einstein Pro - Patent Generation and Document Management Platform [↗](#)

- Backend developer for a platform using AI to generate patents and manage documents.
- Developed a RAG-based chatbot leveraging vector databases for fast and accurate information retrieval.

Money Manager Platform (In Progress)

- Developing a finance management platform with a complete backend built using FastAPI and PostgreSQL for database management [↗](#)
- Created a responsive website using Next.js and a mobile application using Flutter and Dart to provide cross-platform accessibility [↗](#)

Energy Consumption Prediction System (In Progress)

- Designed and deployed an end-to-end energy forecasting pipeline using MLflow for experiment tracking, Apache Airflow for automated data workflows, Docker for containerization, and FastAPI for real-time prediction services.
- Built an interactive Streamlit dashboard to visualize historical consumption, forecasted trends, and key model performance metrics for stakeholder reporting.

PUBLICATIONS

M. S. K. Nisakya, S. M. L. Arampola, W. A. Yasodya, S. Kumarawadu, V. Logeeshan, C. Wanigasekara.
"Self-Adaptive Non-Intrusive Load Monitoring Using Deep Learning." *2024 IEEE World AI IoT Congress*, 2024. [🔗](#) [Best Presented Paper Award]

W. A. Yasodya, S. M. L. Arampola, **M. S. K. Nisakya**, V. Logeeshan, S. Kumarawadu, C. Wanigasekara.
"Self-Adaptive Deep Learning Framework for Non-Intrusive Load Monitoring: Addressing Aging Appliance Challenges With Transfer Learning and Pseudo Labeling." *IEEE Access*, vol. 13, pp. 106524–106539, 2025. [🔗](#)

PROFESSIONAL CERTIFICATIONS

Machine Learning Specialization [🔗](#)
DeepLearning.AI & Stanford University

TensorFlow: Advanced Techniques Specialization [🔗](#)
DeepLearning.AI

Machine Learning Engineering for Production (MLOps) Specialization [🔗](#)
DeepLearning.AI

Data Structures and Algorithms Specialization
University of California San Diego

REFERENCES

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